READ FIRST · UNPACK THE ABBREVIATIONS

RAM (Random Access Memory)

SSD (Solid-State Drive)

COMPUTER SCIENCE / DATA STRUCTURES / YEAR 1

FILES, MEMORY, DATA

Where information “lives” and why it can disappear.

SCAN & LEARN

WHAT HAPPENS • FOLLOW THE ARROWS

01 THE APP IS RUNNING → 02 WORK IS HELD IN RAM → 03 YOU PRESS “SAVE” → 04 THE FILE PERSISTS

FOUR IDEAS TO KEEP

1 RAM ≠ SSD / DRIVE

RAM holds active working data temporarily. SSD/drive preserves files after shutdown. “Save” writes data for later use.

An unsaved document may disappear on shutdown.

2 FILE, FOLDER, PATH

A file holds data stored under a name. A folder groups files. A path tells where a file is inside folders.

Example: Documents / course / assignment.txt.

3 BIT, BYTE, TEXT

A computer stores information as bits (0 or 1). 8 bits make one byte. Text, images and sound are encoded as data.

The letter “A” gets a number using a text encoding.

4 EXTENSIONS AND BACKUPS

.txt, .jpg and .py suggest file types, but do not prove their contents. Backups protect your work.

“In the cloud” usually means on another computer (server).

TRY IT — DRAW A FILE PATH

On paper: university → Year 1 → lab → task.py. Which is the file? Which are folders?

THE ONE SENTENCE TO REMEMBER

RAM = worktable · SSD = storage cabinet · FILE = saved item.

Question: What do you risk if you shut down without saving?